

SMITHIAN FOLD THEORY V3

Twelve Verdicts From the Fold

Strict SFT proof/disproof of OpenAI's ten advances in mathematics and theoretical computer science

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Publication control. <i>This is a new versioned successor. It does not rewrite the preceding paper, its DOI record, historical receipts or evidence. No push, upload, DOI action, release or publication is authorised until Maria Smith explicitly confirms the exact final files and hashes.</i>
Lean verification integration. <i>The current SFT ordered census passed the independent Lean 4 verification layer on 2 August 2026: 2,777 accepted claims, 898,902 candidate records and decisions, 11,108 controls, seventeen branches and no reported issue. Lean natively proves the two-class operational root and its unique survivor and constructs proof-bearing acceptance-gate certificates. The remaining claim content is checked as the complete registered artifact graph; that distinction is retained throughout this successor.</i>

Abstract

This successor preserves the preceding conclusion-level and admissibility analysis and adds the methodological consequence of SFT's own Lean 4 verification, which passed all 2,777 claims in the current ordered census. Formal acceptance establishes the encoded proposition; SFT admission additionally binds that proposition to its generated grammar, ownership, evidence boundary, candidate decisions, controls and receipts. The new verification does not reclassify any OpenAI comparison verdict or turn external Lean acceptance into automatic SFT theoremhood.

OpenAI announced ten advances in mathematics and theoretical computer science on 1 August 2026 and released manuscripts, reasoning walkthroughs and Lean 4 formalizations. This paper answers a narrower and more demanding question than whether Lean accepts those formalizations: do the exact mathematical conclusions hold when they are translated without weakening into the already existing mathematical language and constraints of Smithian Fold Theory (SFT)?

The official release is frozen locally at commit `94bc0feb6a9ff12c7d31d6de640a725c9d43d2b6`. The ten advertised advances resolve into twelve main Lean declarations. Each declaration is assigned to exactly one categorical SFT owner: nine to Mathematics, two to Classical Computation and one to Quantum Computation. For every declaration, the target is its exact conclusion, not the admissibility of its submitted Lean proof.

The strict translation rule preserves quantifiers, carrier types, operators, comparisons, resource measures and generality. A real limsup is not replaced by a finite table; a complex fraction-ring formula is not renamed a Fold circuit; an infinite

group is not replaced by a large finite group. Under that rule, an exact conclusion holds in SFT only if every logically necessary component is an admitted SFT object or operation. Existing model-admitted SFT results deny completed infinity, ungenerated continua, unrestricted real and complex proof scalars, completed at-infinity filters, unregistered all-problem families and untranslated resource models. Each OpenAI conclusion contains at least one such necessary component. Assuming any exact conclusion holds in SFT therefore implies both admission and denial of the same required component. By contradiction, every exact conclusion is denied inside SFT.

One verdict is stronger than a carrier contradiction. For every SFT group, the exact left-regular action is a finite permutation model: multiplication is preserved with empty disagreement support, while every nonidentity element moves every carrier point and has normalised separation One. Hence every SFT group is sofic and no SFT non-sofic witness exists. The other eleven are structural disproofs: their exact proposition cannot be inhabited or asserted in the SFT model because a required field, carrier, operator or resource object is expressly denied by existing admitted law.

Formal admission, implementation validation, observation, measurement, empirical support, confirmation, adverse evidence, unresolved evidence and publication status remain separate classifications. Lean does not reclassify an empirical result, repair an adverse observation or convert a later comparison into an earlier prediction. Lawful criticism, falsification and versioned extension remain open.

Successor headline findings

1. **Independent whole-model result.** Lean 4 returned `PASS` for 2,777/2,777 current claims, 898,902 candidates and decisions, 11,108 controls and all 17 branches, with no issue.
2. **Operational root.** The exact registered two-class grammar is formalised natively, and `presentedOccurrence` is proved to be its unique survivor without an imported or user-declared axiom in the exported theorem's axiom audit.
3. **Typed validation.** The root and generic acceptance implications are native Lean theorems; the remaining scientific claims are checked as complete registered artifacts. Empirical truth is not inferred from proof-assistant acceptance alone.
4. **Fail-closed custody.** A six-file byte-identity mismatch halted the first run. Exact registered bytes were restored, all 385 source bindings were re-audited, and the unchanged verifier then passed.

These findings supplement rather than erase the branch-specific headline results retained below. Any adverse, corrected, unresolved, unavailable or chronology-bound result in the preceding scientific record remains governed by its current claim package.

Current status, evidence language and reader map

Status field	Current position
Paper and completion boundary	methodological or comparison scope; no downstream claim ownership transferred. Completion is dated to the current registered census and remains open to lawful versioned extension.
Formal status	The Lean root theorem is kernel checked. Every live model claim has a current admitted receipt and passed the Lean artifact gates. Formal admission does not imply empirical confirmation.
Empirical status	Claim specific. Blind, non-blind, development-observed, holdout, favourable, adverse, null, missing, unavailable, disputed and unresolved distinctions remain exactly as recorded.
Chronology	Claim-specific registration, sealing, custody and observation order controls. Later evidence never becomes an earlier prediction without the registered chronology.
Publication status	Version 0.2.0 is a final local publication candidate awaiting Maria Smith's approval. It is not deposited or published.
Ownership	This paper retains its declared scientific boundary. Lean verification creates no new branch ownership and no application may select a law.
What is not claimed	Permanent closure of science, universal empirical confirmation, conversion of compatibility into confirmation, or superiority to every rival model.

Corpus-wide terminology and evidence key

Term	Reserved meaning in this paper
Theorem	A formally closed proposition at its declared grammar and dependency boundary.
Law	An admitted branch relation with its declared carrier, boundary, dependencies and extension rule.
Claim	The registered unit judged by the engine and bound to one immutable receipt.
Constitution	The rules governing admissible objects, derivations, evidence and publication; not an empirical result.
Derivation	Generation and elimination of the declared candidate space to its surviving structure.
Prediction	A consequence sealed before the matching target is released under the registered custody protocol.
Observation	A source-bound external record; it does not enter candidate generation.
Measurement	An instrument-, method-, condition- and uncertainty-bound observed value.
Reconstruction	A separately implemented regeneration or an explicitly identified inference of a retained state or history.
Exact numerical correspondence	Equality or registered interval relation between exact numerical objects at the declared boundary.
Structural correspondence	A post-derivation relation between forms without a claim of exact numerical prediction.
Boundary correspondence	Agreement restricted to the named interface, limit or ownership boundary.
Compatibility	Non-adverse but non-discriminating evidence.
Support	Relevant evidence that is not unique confirmation.
Confirmation	Used only when the current registered evidence protocol warrants that classification.
Validation	Successful execution of the named formal, computational or empirical test; its kind must be stated.
Adverse result	A registered test result that conflicts with the tested claim at its declared boundary.
Unresolved result	Required evidence remains unavailable, incomplete, disputed or insufficiently classified.
Implementation identity	The hash-bound identity of executable material; implementation success is not empirical confirmation.
Formal, empirical and publication status	Three independent classifications; none substitutes for another.
Foundational closure	Completion of the registered foundation only; not field-wide closure.
Field-wide closure	Completion of the named frozen or registered dated field census.
Current-evidence closure	Closure only to the evidence surface presently registered and preserved.
Extension openness	New questions may enter a later version without rewriting receipts or history.

Proof language is reserved for formal closure; derivation for generated structure; implementation for executable demonstration; prediction for a correctly sealed prospective consequence; observation and measurement for source-bound external records; reconstruction for separately regenerated or inferred states; correspondence for post-derivation relations; compatibility and support for non-unique evidence; confirmation only where the registered protocol authorises it; adverse for conflict; and unresolved where required evidence remains incomplete.

Three reading levels

1. **Conceptual paper.** The abstract, headline findings, branch narrative, major derivations, evidence synthesis, limitations and conclusion provide the readable scientific argument.
2. **Scientific audit layer.** Family and claim sections preserve exact status, dependencies, candidate and survivor counts, controls, chronology, sources, corrections, adverse evidence and reconciliation.
3. **Machine archive.** The repository packages preserve complete candidates, decisions, hashes, receipts, executable traces, source snapshots and certificates. They remain authoritative where prose abbreviates display.

Editorial change control

This successor improves currency, expression, typography and navigation. It does not change scientific meaning, chronology, evidence class, branch ownership, claim status or machine identity. Any conflict between authoritative records must be resolved by explicit supersession or reported to Maria Smith; prose alone cannot manufacture agreement.

1. The question answered

OpenAI's public claim and this paper's target must not be conflated. OpenAI presents ten results and a Lean certificate for each. The present paper asks:

For each exact main conclusion, does the conclusion survive a strict, structure-preserving translation into the existing SFT model?

The answer is no in all twelve atomic cases.

Let C_i denote the exact conclusion of atomic result i . Let $\text{Holds_SFT}(C_i)$ mean that the strict translation of C_i is a lawful, inhabited conclusion of SFT. Let $\text{Req}(C_i)$ be the set of carrier types, operations, quantifier ranges, comparison objects and resource measures logically required by the exact statement. Let $\text{Adm_SFT}(r)$ mean that required component r is admitted by SFT.

Strict conclusionhood gives the following necessary rule:

$\text{Holds_SFT}(C_i) \rightarrow \text{for every } r \text{ in } \text{Req}(C_i), \text{Adm_SFT}(r).$

For every result in the source set, source inspection identifies at least one required component r_i and existing model-admitted SFT evidence gives:

$\text{not Adm_SFT}(r_i).$

Therefore:

Assume $\text{Holds_SFT}(C_i).$
 Then $\text{Adm_SFT}(r_i).$ [strict conclusionhood]
 But $\text{not Adm_SFT}(r_i).$ [existing SFT result]
 Contradiction.
 Therefore $\text{not Holds_SFT}(C_i).$ [SFT disproof]

This rule does not use a missing SFT receipt for an OpenAI artifact as evidence of falsity. It does not use OpenAI's imported Lean axioms as a conclusion-level premise. It uses the exact source statement plus positive, already admitted SFT results that force the opposite orientation for a necessary component.

1.1 Meaning of the verdict word

In this paper:

- **PROVED** means the complete strict SFT translation has a positive, receipt-backed SFT derivation.
- **DISPROVED** means an existing model-admitted SFT result forces denial of a logically necessary component, so the complete strict SFT conclusion cannot hold.
- **OPEN** is not an allowed output for this closed comparison surface.

The conclusion-level verdict is different from the earlier artifact-admissibility audit. Rejecting a Lean artifact because it relies on a foreign proof stack does not by itself refute its conclusion. This paper instead identifies the exact necessary content of every conclusion and derives its denial from SFT's existing object and operation laws.

1.2 Why no weakening is allowed

Weakening would manufacture agreement. Replacing an all-dimension real limit by a finite list of checked dimensions, a complex formula by a Fold circuit, or an infinite group by a finite table changes the proposition. Such finite SFT analogues may be meaningful, and some are already admitted, but they do not prove the exact OpenAI statement. The strict test therefore preserves:

- every existential and universal quantifier;
- each carrier, including real, complex, set, group and strategy carriers;
- each operator, including $\limsup$, $\inf$, $\sup$, Tendsto , logarithm and real exponentiation;
- each resource measure and reduction family; and
- the stated range of dimensions, code levels, languages, groups, games or graphs.

If preservation exposes a direct conflict with SFT, the result is DISPROVED in SFT. Rephrasing it until the conflict disappears is not permitted.

2. Frozen source and exact atomic targets

The official OpenAI sources were captured before analysis. The fixed source commit is 94bc0feb6a9ff12c7d31d6de640a725c9d43d2b6. The local custody record stores hashes, sizes, origin URLs, rights and quarantine status. Source line numbers below refer to that commit.

Atomic ID	Advance	Exact main declaration and fixed location	Owner
OAI26-MATH-001	1	PackingBounds.sharpFullCohnE lkiesManuscriptConclusions, SpherePacking.lean:55591	Mathematics
OAI26-MATH-002	2	MetricCodes.Johnson.binaryRa te_lt_mrrw, MetricCodes.lean:20797	Mathematics
OAI26-MATH-003	2	MetricCodes.Spherical.Higher Hierarchy.strict_hierarchy, MetricCodes.lean:114336	Mathematics
OAI26-MATH-004	3	SoficGroups.SourceTopLevelCo mpressionFinal.exists_finite lyPresented_nonsofic_group, NonSoficGroup.lean:34430	Mathematics
OAI26-MATH-005	4	ConnesRigidity.exists_infini te_pairwise_nonisomorphic_pr opertyT_icc_groups_with_isom orphic_factors, ConnesRigidity.lean:37348	Mathematics
OAI26-COMP-001	5	PermanentFormulaLowerBound.p ermanent_rational_formula_lo garithmic_lower_bound, Permanent.lean:27551	Classical Computation
OAI26-QUANTUM-001	6	QuantumParallelRepetition.di stributionUniformExponential, QuantumParallelRepetition.le an:70953	Quantum Computation
OAI26-COMP-002	7	GapCVP.Comparator.gapCVP400I sNPHard, GapCVP.lean:130398	Classical Computation
OAI26-MATH-006	8	Ehrhart.Volume.ehrhart_volum e_inequality_for_sets, Ehrhart VolumeInequality.lean:55733	Mathematics
OAI26-MATH-007	9	ErdosProblems.MulticolourTri angleRamsey.erdos_problem_18 3_explicit, MulticolorTriangle Ramsey.lean:3042	Mathematics
OAI26-MATH-008	10	CompactnessConjecture.quantit ativeCompactnessCounterexam ple, CompactnessAndDegeneracy. lean:9320	Mathematics
OAI26-MATH-009	10	TwoDegenerateGraphs.twoDegen erateExtremalCounterexample, C ompactnessAndDegeneracy.lean :18441	Mathematics

The ten public advances become twelve atomic targets because the coding result has separate binary and spherical declarations, while the extremal-graph result has separate compactness and two-degeneracy declarations. A grouped advance is denied if any required atomic conclusion is denied; here every atomic conclusion is denied.

3. Existing SFT constraints used in the proofs

No new SFT law is introduced to obtain these verdicts. The proof chains consume already registered, certified and model-admitted claims. The central constraints are these.

3.1 Generated finite succession, not completed infinity

SFT-FOUNDATION-COUNT-001 gives every generated nonempty finite succession one exact count representation.

SFT-MATH-LIMIT-CONTINUUM-002 states that refinement may continue without a greatest reached depth, while every reached object remains finite and exact. It expressly places completed infinity, an unbounded denominator and an ungenerated continuum outside the admitted object language.

Thus SFT distinguishes potential continuation from a completed infinite object. `atTop`, an all-natural completed family, an arbitrary real set and an infinite carrier are not silently identified with a long finite census.

3.2 Exact positive operations

SFT-FOUNDATION-EXACT-OPERATIONS-001 admits exact positive rational parts through the One, exact cast and Fold, guarded Take, unison and root-bound trace. It does not admit negative, imaginary or unrestricted irrational proof quantities as native scalars. Exact squared separation may carry a distance comparison when a conventional square root would be irrational, under SFT-MATH-GEOM-EUCLIDEAN-DISTANCE-002.

3.3 Convergence by finite rational witness

SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006 defines convergence through nested exact rational enclosures whose widths refine under a certified rule. SFT-MATH-MEAS-CONVERGENCE-FINITE-WITNESS-010 requires exact finite witness widths at every registered successor and keeps completed infinite equality outside the certificate.

These laws permit exact successor-indexed evidence. They deny completed real `limsup`, `sInf`, `sSup` and `Tendsto` ... `atTop` objects when those objects are required as completed proof scalars rather than represented by a finite witness law.

3.4 Finite exact mathematics

The relevant Mathematics claims admit:

- exact finite code maxima from complete word-subset censuses (SFT-MATH-COMB-CODING-PACKING-010);
- packing, covering and tessellation at declared finite boundaries (SFT-MATH-GEOM-PACKING-COVERING-TESELLATION-015);
- groups through exact held inverses and reversible finite actions (SFT-MATH-ALG-GROUP-HELD-INVERSE-004, SFT-MATH-ALG-PERMUTATION-GROUP-ACTION-005);
- finite operator image support and finite-support integration (SFT-MATH-ANAL-BOUNDED-COMPACT-OPERATOR-008, SFT-MATH-MEAS-FINITE-SUPPORT-INTEGRATION-005);
- exact finite convex hull, lattice-polytope and incidence custody (SFT-MATH-GEOM-CONVEX-HULL-SEPARATION-005, SFT-MATH-GEOM-DISCRETE-LATTICE-POLYTOPE-006); and
- exhaustive finite Ramsey, coloring and extremal censuses (SFT-MATH-COMB-RAMSEY-FORCING-011, SFT-MATH-GRAPH-COLOURING-CONSTRAINT-006, SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007, SFT-MATH-GRAPH-MATCHING-COVERING-PACKING-007).

The words finite and exact are substantive constraints. They do not mean that a completed continuum statement inherits authority because every requested finite sample could in principle be generated.

3.5 Computation and quantum computation

SFT-COMP-CPLXX-FORMULA-BRANCHING-CIRCUIT-014 permits formula/circuit correspondence only through translations preserving input support, output labels, sharing records and exact overhead.

SFT-COMP-CPLXX-ARBITRARY-FOLD-CIRCUIT-LOWER-024 is explicitly a lower bound for circuits assembled from admitted Fold edges; other gate bases require a separate transport.

SFT-COMP-CPLXX-REDUCTION-COMPLETE-PROBLEM-021 requires every registered family member to have a total verdict-preserving reduction with explicit compositional overhead.

SFT-QUANTUM-ENTANGLEMENT-001 admits exact finite nonfactorable joint-word support and explicitly imports no complex amplitudes or postulates. SFT-QUANTUM-QCPLXX-PARALLEL-016 retains work, depth, processors, width, live

support and communication as separate finite resource coordinates. These are not aliases for complex Hilbert space, a supremum over all strategies or a real exponential error curve.

4. Complete translation-verdict enumeration

The proof report is a downstream comparison object. It does not edit the immutable SFT engine, add a claim to the model or treat OpenAI's answers as derivational authority.

For each atomic target, the audit enumerates the complete Cartesian product of six binary dimensions:

Dimension	Rejected coordinate	Required surviving coordinate
Scope	artifact admissibility	mathematical conclusion
Translation	weakened or replaced	strict preserving
Necessary component	all components native	SFT-denied component present
Evidence	missing or nonauthoritative	existing model-admitted negation
Orientation	positive	held denial
Verdict	PROVED	DISPROVED

There are $2^6 = 64$ forms per atomic conclusion and $12 \times 64 = 768$ forms overall. Exactly one coordinate survives for each result:

```

mathematical conclusion
+ strict preserving translation
+ SFT-denied necessary component present
+ existing model-admitted denial evidence
+ held denial orientation
= DISPROVED

```

The enumeration does not make its own semantic premises true. Its role is narrower and auditable: it prevents a shift back to artifact admissibility, a weakened finite analogue, missing-evidence reasoning, a positive orientation unsupported by the dependency trace, or a nonbinary verdict. The semantic work is done by the exact source component and the cited admitted SFT result in each proof below.

5. Closed verdict table

Atomic ID	Exact conclusion	Decisive SFT contradiction	Verdict
OAI26-MATH-001	Sharp Cohn-Elkies asymptotics	Completed real limits and real-valued error functions versus finite rational witness law	DISPROVED
OAI26-MATH-002	Binary rate strictly below MRRW	Real $\limsup/s\text{Inf}$ rate objects versus exact finite code census and rational enclosure	DISPROVED
OAI26-MATH-003	Strict spherical-code hierarchy	Completed all-level real hierarchy versus generated finite levels	DISPROVED
OAI26-MATH-004	Finitely presented non-sofic group exists	Exact left-regular action makes every finite SFT group sofic	DISPROVED
OAI26-MATH-005	Infinite property-(T) ICC family with isomorphic factors	IsICC requires $\text{Infinite } G$; SFT group carriers are generated finite	DISPROVED
OAI26-COMP-001	Permanent formula lower bound	Complex fraction-ring formula and real-log resource versus native positive Fold operations and untranslated gate basis	DISPROVED
OAI26-QUANTUM-001	Uniform exponential quantum repetition	Complex density/POVM strategies and real supremum/exponential versus finite word-support quantum law	DISPROVED
OAI26-COMP-002	GapCVP is NP-hard at polynomial factor	All-language conventional hardness and signed/real gap carrier versus registered finite reductions and exact distance	DISPROVED
OAI26-MATH-006	Sharp Ehrhart volume inequality	Arbitrary real sets, topology and continuum volume versus finite hull and finite-support integration	DISPROVED
OAI26-MATH-007	Explicit Ramsey lower bound and divergence	Required completed $\text{Tendsto} \dots \text{atTop}$ conjunct versus finite Ramsey census law	DISPROVED
OAI26-MATH-008	Compactness counterexample package	Required eventually-atTop lower witness versus finite graph census and limit law	DISPROVED
OAI26-MATH-009	Two-degenerate extremal counterexample	Required real epsilon/exponent and eventually-atTop witness versus exact finite graph law	DISPROVED

6. Twelve source-level SFT proofs and disproofs

6.1 Advance 1: high-dimensional sphere packing

Target. `PackingBounds.sharpFullCohnElkiesManuscriptConclusions` inhabits a proposition-valued structure with ten required fields. Among them are several `Tendsto` statements at `atTop`, `real sInf`, `Real.sqrt`, `Real.exp`, `Real.pi`, real logarithms and three existential functions of type `Nat → Real` whose values vanish at infinity.

Necessary component. An inhabitant of the complete structure must inhabit every field. In particular it must provide a completed real-valued error function and a completed at-infinity convergence relation. These are not optional proof conveniences; they are fields of the conclusion itself.

SFT premises. `SFT-MATH-LIMIT-CONTINUUM-002` permits a never-terminal successor process but denies completed infinity and an ungenerated continuum as proof objects.

`SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006` admits convergence only through exact nested rational enclosures. `SFT-FOUNDATION-EXACT-OPERATIONS-001` restricts proof scalars to the exact positive domain.

`SFT-MATH-ANAL-HARMONIC-FOURIER-SUPPORT-009` supplies a finite exact harmonic correspondence without imaginary proof scalars.

Derivation. Assume the exact record holds in SFT. Then its required `root_before_infimum` field admits a completed real `sInf` sequence and `Tendsto ... atTop`; its vanishing-error fields admit completed functions `Nat → Real`. Strict preservation therefore yields admission of completed continuum-valued limit objects. The cited SFT limit law denies those same objects. Contradiction. Because a structure is inhabited only if every required field is inhabited, one denied field denies the whole conclusion.

Verdict OAI26-MATH-001: DISPROVED in SFT. The exact asymptotic record has no SFT inhabitant. Finite packing and rational enclosure results are different, weaker propositions and do not alter this verdict.

6.2 Advance 2A: strict binary-code improvement over MRRW

Target. `MetricCodes.Johnson.binaryRate_lt_mrrw` states, for real d with $0 < d < 1/2$, that `binaryRate d < mrrwRate d`. In the fixed source, `binaryRate` is a real `Filter.limsup` of normalised logarithms over `Filter.atTop`; `mrrwRate` is the real `sInf` of a real variational set and its construction uses logarithmic and square-root operations.

Necessary component. The strict inequality compares two completed real asymptotic-rate scalars. It is not merely a family of exact inequalities for each finite block length.

SFT premises. `SFT-MATH-COMB-CODING-PACKING-010` admits the exact maximum of a complete finite word-subset census. `SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006` admits certified nested rational bounds. `SFT-MATH-LIMIT-CONTINUUM-002` denies a completed real limsup and ungenerated continuum as proof objects.

Derivation. Assume the exact strict inequality holds in SFT. Because strict translation preserves both sides, SFT must admit `binaryRate d` as the completed real limsup and `mrrwRate d` as the completed real infimum for the stated real domain. The SFT limit law denies those completed rate objects and instead requires finite generated values with enclosure certificates. Thus the assumption entails both admission and denial of a necessary comparison scalar. Contradiction.

***Verdict OAI26-MATH-002: DISPROVED in SFT.** SFT can compare registered finite code maxima and certified enclosure bounds, but that is not the exact real limsup-versus-infimum proposition.*

6.3 Advance 2B: strict spherical-code hierarchy

Target. `MetricCodes.Spherical.HigherHierarchy.strict_hierarchy` asserts, for real s in $(0, 1)$, strict descent for every natural level r , followed by a chain of inequalities between `sphericalCodeRate`, `localizedHierarchyRate`, localized level rates and a classical localized rate. The source defines level rates by real `sInf`, the spherical-code rate by real `limsup` at `atTop`, and the hierarchy rate by the infimum of the range over all natural levels.

Necessary component. The theorem requires one completed all-level real hierarchy, not any chosen finite prefix.

SFT premises. `SFT-MATH-COMB-CODING-PACKING-010` and `SFT-MATH-GEOM-PACKING-COVERING-TESELLATION-015` admit exact finite code and packing structures. `SFT-MATH-LIMIT-CONTINUUM-002` denies the completed ungenerated hierarchy and completed real limit carrier.

Derivation. Assume the exact hierarchy holds in SFT. The first conjunct admits a total all-natural strict hierarchy; the later conjuncts admit completed real infima and a real limsup rate. Strict preservation forbids truncating r or replacing those rates with finite tables. The assumption therefore admits the completed hierarchy object denied by the SFT limit law. Contradiction.

***Verdict OAI26-MATH-003: DISPROVED in SFT.** Every generated finite level can remain a lawful object; their completed real range and infimum cannot.*

6.4 Advance 3: existence of a non-sofic group

Target.

`SoficGroups.SourceTopLevelCompressionFinal.exists_finitelyPresented_nonsofic_group` asserts the existence of a type G with a group structure, a finite presentation and not `Sofic G`.

This target receives a direct SFT counterproof rather than only a carrier objection.

SFT premises. By the generated-carrier and limit laws, every SFT group carrier is finite.

`SFT-MATH-ALG-GROUP-HELD-INVERSE-004` supplies exact identity and inverse closure.

`SFT-MATH-ALG-PERUTATION-GROUP-ACTION-005` admits reversible carrier-preserving actions.

Construction. Let G be any SFT group. For each g in G , define the left-regular permutation

$$L_g : G \rightarrow G, \quad L_g(x) = g * x.$$

It is reversible with inverse $L_{(g^{-1})}$. For all g, h, x :

$$L_{(g * h)}(x) = (g * h) * x = g * (h * x) = (L_g \circ L_h)(x).$$

Therefore multiplication is preserved exactly. The disagreement support between $L_{(g * h)}$ and $L_g \circ L_h$ is Empty.

Now take $g \neq 1$. If some x were fixed by L_g , then $g*x = x$. Composition with the held inverse of x gives $g = 1$, a contradiction. Hence every x is moved. The moved support is the complete carrier and its normalised Hamming separation is One.

For any generated finite test set F and any exact positive tolerance ϵ below One, choose the permutation model with carrier G and action $g \mapsto L_g$. Its multiplicative disagreement is Empty, hence below ϵ ; every nonidentity element has separation One, hence above ϵ . Take ϵ . This satisfies the exact finite permutation-approximation conditions.

Conclusion. Every SFT group is sofic. Therefore:

```
not exists G, SFTGroup(G) and not Sofic(G).
```

The additional requirement of finite presentation cannot restore a witness after the non-sofic conjunct has been denied.

Verdict OAI26-MATH-004: DISPROVED in SFT. The left-regular action is a universal SFT counterproof to the requested non-sofic witness.

6.5 Advance 4: Connes rigidity counterexample

Target. `ConnesRigidity.exists_infinite_pairwise_nonisomorphic_propertyT_icc_groups_with_isomorphic_factors` asserts an existential natural-indexed family of countable discrete groups. Every member and a distinguished group must be ICC and have property (T); their tracial group factors must be isomorphic; the groups must be pairwise nonisomorphic.

Decisive source fact. In the fixed Lean source:

```
IsICC(G) := Infinite(G)
and every nonidentity conjugacy class is infinite.
```

Thus a completed infinite carrier is a literal positive conjunct of every requested witness.

SFT premises. SFT group carriers and operator supports are generated finite objects under

SFT-MATH-ALG-GROUP-HELD-INVERSE-004,

SFT-MATH-ALG-REPRESENTATION-ACTION-DECOMPOSITION-013 and

SFT-MATH-ANAL-BOUNDED-COMPACT-OPERATOR-008. Integration is finite-support under

SFT-MATH-MEAS-FINITE-SUPPORT-INTEGRATION-005. SFT-MATH-LIMIT-CONTINUUM-002 denies a completed infinite carrier and an all-natural completed family.

Derivation. Assume the exact existential conclusion holds in SFT. Its `IsICC` `Lambda` field entails `Infinite` `Lambda`. But every admitted SFT group carrier is generated and finite; completed `Infinite` `Lambda` is denied. This already contradicts a necessary witness field. Independently, the exact conclusion also requires an all-natural completed family and complex Hilbert/operator factors, which conflict with the finite-support operator laws. One failed required field is sufficient to deny the existential conjunction.

Verdict OAI26-MATH-005: DISPROVED in SFT. No SFT witness can satisfy the explicit `Infinite` conjunct of `IsICC`.

6.6 Advance 5: permanent lower bounds for rational formulas

Target. `PermanentFormulaLowerBound.permanent_rational_formula_logarithmic_lower_bound` quantifies over a `RationalFormula` whose variables and values lie in the complex field. Validity and evaluation occur in the fraction ring of the complex multivariate-polynomial ring. The conclusion compares the number of variable leaves with the real scalar $n^4 / (192 \log_2 n)$.

Necessary component. The exact statement fixes both a conventional complex fraction-ring gate model and a real-logarithmic resource comparison. Replacing either changes the theorem.

SFT premises. SFT-FOUNDATION-EXACT-OPERATIONS-001 admits exact positive operations and denies imaginary and unrestricted negative proof quantities. SFT-MATH-SYMB-CANONICAL-EXPRESSION-001 identifies expressions only through registered canonical syntax rules. SFT-COMP-CPLXX-FORMULA-BRANCHING-CIRCUIT-014 requires a trace-preserving translation between computational models.

SFT-COMP-CPLXX-ARBITRARY-FOLD-CIRCUIT-LOWER-024 expressly confines its current lower-bound result to circuits assembled from admitted Fold edges and says that other gate bases require separate transport.

Derivation. Assume the exact lower-bound conclusion holds in SFT. Strict preservation requires the complex field, the fraction-ring formula semantics, subtraction/division behavior and the real logarithmic resource scalar to be native, or requires an already admitted exact transport into the Fold resource model. The exact-operation law denies the first route. The formula/circuit and lower-bound laws deny identification without a preserving transport, and the existing Fold lower-bound claim explicitly does not supply one for this gate basis. Hence the assumption forces an exact carrier-and-resource identification that existing SFT law denies. Contradiction.

This is not an argument from an absent external receipt. The positive SFT lower-bound law states its own closure boundary: Fold edges only, with other bases requiring transport. Treating a complex rational formula as already transported would contradict that admitted result.

Verdict OAI26-COMP-001: DISPROVED in SFT. *The exact complex fraction-ring proposition is incompatible with the admitted SFT carrier and resource model.*

6.7 Advance 6: quantum parallel repetition

Target. `QuantumParallelRepetition.distributionUniformExponential` asserts the existence of a positive real constant c that uniformly bounds repeated entangled values for every finite two-player game and every positive repetition count by a negative real exponential.

The source defines a strategy through finite-dimensional complex density matrices and complex POVMs. `entangledValue` is the real supremum of the range of all such strategy win probabilities. The bound then uses subtraction from One, real logarithm, real division, a thirteenth power and `Real.exp` of a negative real product.

Necessary component. The theorem requires the conventional complex strategy carrier and its completed real supremum before its exponential bound is even formed.

SFT premises. `SFT-QUANTUM-ENTANGLEMENT-001` defines entanglement as exact finite nonfactorable joint-word support and explicitly imports no complex amplitudes or Hilbert-space postulate. `SFT-QUANTUM-QCPLXX-PARALLEL-016` retains a finite vector of work, depth, processors, register width, live support and communication.

`SFT-FOUNDATION-EXACT-OPERATIONS-001` denies the required negative and unrestricted real proof scalars.

Derivation. Assume the exact uniform exponential conclusion holds in SFT. Then the source-defined `entangledValue` G must be an admitted real `sSup` over all complex density-matrix/POVM strategies, and the right side must be an admitted negative real exponential. The SFT entanglement claim expressly denies imported complex amplitudes and replaces them with exact finite joint-word support; the SFT resource claim preserves a finite resource vector rather than a completed real strategy supremum. Strict translation prohibits substituting that different quantum object. The assumption therefore admits precisely the complex/supremum carrier that existing SFT law denies. Contradiction.

Verdict OAI26-QUANTUM-001: DISPROVED in SFT. *A finite exact SFT repetition law would be a different theorem, not a proof of this complex-supremum exponential statement.*

6.8 Advance 7: closest-vector hardness

Target. `GapCVP.Comparator.gapCVP400IsNPHard` asserts `IsNPHardPromise gapCVP400Promise`. In the fixed source, `IsNPHardPromise` decides a proposition quantifying over every bit language: whenever a language is in NP, there must be a nonempty polynomial-time promise reduction to the target. The target representation uses conventional signed integer lattices, Euclidean norms and a real polynomial gap.

Necessary component. The exact claim requires a completed all-bit-language reduction family and the conventional signed/real target semantics. A finite registered list of reductions is not the same proposition.

SFT premises. `SFT-COMP-CPLXX-REDUCTION-COMPLETE-PROBLEM-021` calls a problem complete only for a declared family whose every registered member has a total verdict-preserving reduction with explicit compositional overhead. `SFT-COMP-CPLXX-APPROXIMATION-RATIO-026` requires an exact candidate-to-optimum part with declared orientation and no hidden oracle. `SFT-MATH-GEOM-DISCRETE-LATTICE-POLYTOPE-006` admits finite exact lattice carriers, while `SFT-MATH-GEOM-EUCLIDEAN-DISTANCE-002` carries distance through exact squared separation when a root would be irrational. The exact-operation law denies unrestricted signed and irrational proof scalars.

Derivation. Assume the exact `IsNPHardPromise` conclusion holds in SFT. Then one completed proposition ranges over every bit language and imports a conventional polynomial-time reduction authority into a signed-integer, real-gap target. The SFT completeness law admits only a declared registered family with explicit total maps and full resource custody. The SFT

lattice and distance laws do not identify their finite exact carriers with the conventional signed/irrational target without transport. Thus the assumption requires an unregistered completed family and untranslated target semantics expressly excluded by the existing SFT reduction boundary. Contradiction.

Verdict OAI26-COMP-002: DISPROVED in SFT. The exact all-language conventional NP-hardness proposition is denied; this does not weaken it into a finite registered reduction result.

6.9 Advance 8: Ehrhart volume inequality

Target. `Ehrhart.Volume.ehrhart_volume_inequality_for_sets` ranges over an arbitrary set S in a real vector space. Its hypotheses use real convexity, topological compactness, nonempty topological interior, a numerically zero barycenter and a singleton zero interior-lattice-point set. Its conclusion compares normalised continuum volume with a real factorial expression.

Necessary component. The exact proposition requires an arbitrary real-continuum set, its topology and its continuum volume. It is not merely a finite lattice-polytope inequality.

SFT premises. `SFT-MATH-GEOM-CONVEX-HULL-SEPARATION-005` admits the finite exact hull of lawful generated combinations. `SFT-MATH-GEOM-DISCRETE-LATTICE-POLYTOPE-006` defines a lattice polytope as the finite hull of exact coordinate carriers with boundary and interior custody. `SFT-MATH-MEAS-FINITE-SUPPORT-INTEGRATION-005` admits exact integration over finite support; `SFT-MATH-MEAS-CONVERGENCE-FINITE-WITNESS-010` does not identify this with completed continuum equality. The exact-operation law denies numerical zero and an unrestricted real proof domain.

Derivation. Assume the exact set theorem holds in SFT. Strict preservation requires SFT to admit arbitrary subsets of a real continuum, their topological interior and compactness, Lebesgue-style normalized volume and numerical-zero barycenter conditions. The cited SFT geometry and measure laws instead admit finite generated hulls and finite-support accumulation and deny the continuum/numerical-zero carrier. The assumption therefore entails admission and denial of a necessary set-and-volume object. Contradiction.

Verdict OAI26-MATH-006: DISPROVED in SFT. A finite exact lattice-polytope enclosure theorem would be a replacement, not the exact conclusion.

6.10 Advance 9: multicolour triangle Ramsey numbers

Target. `ErdosProblems.MulticolourTriangleRamsey.erdos_problem_183_explicit` is a conjunction. Its first conjunct gives, for every natural $k \geq 2$, a real lower bound involving `Real.exp`, `Real.log` and fractional real powers. Its second conjunct states:

`Tendsto (k |-> triangleRamseyNumber(k)^(1/k)) atTop atTop.`

Necessary component. Both conjuncts are required. In particular, the completed at-infinity divergence statement cannot be dropped while retaining the exact theorem.

SFT premises. `SFT-MATH-COMB-RAMSEY-FORCING-011` closes a Ramsey boundary only through the complete declared finite colouring census and a preceding counterexample where claimed.

`SFT-MATH-GRAPH-COLOURING-CONSTRAINT-006` and `SFT-MATH-COMB-CODING-PACKING-010` likewise operate on exact finite candidates. `SFT-MATH-LIMIT-CONTINUUM-002` denies the completed `atTop` object.

Derivation. Assume the exact conjunction holds in SFT. Then its second conjunct admits a completed `Tendsto ... atTop atTop` relation. The SFT limit law denies that relation as a completed proof object and permits only successor-generated finite evidence. Contradiction. Because $A \text{ and } B$ requires both A and B , denial of B denies the exact conjunction regardless of the status of any individual finite Ramsey inequality.

Verdict OAI26-MATH-007: DISPROVED in SFT. Finite Ramsey censuses survive as separate propositions; the required completed divergence conjunct does not.

6.11 Advance 10A: compactness counterexample

Target. `CompactnessConjecture.quantitativeCompactnessCounterexample` asserts an existential package containing a finite nonempty forbidden family and positive real constants, structural graph fields, `UniformMemberLower`, real power inequalities for every host size, not `IsCompactFamily` and not `CompactnessConjectureStatement`.

The source defines `UniformMemberLower` with an eventually-at-atTop lower bound for every family member. It defines `IsCompactFamily` through an existential positive real constant and another eventually-at-atTop comparison. Thus completed eventual asymptotics are part of the witness package and the predicate it negates.

Necessary component. The positive `UniformMemberLower` field alone requires a completed eventually-at-infinity real comparison. Its omission would destroy the exact existential witness.

SFT premises. `SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007` and `SFT-MATH-GRAPH-MATCHING-COVERING-PACKING-007` admit exhaustive finite feasible-family and graph censuses. `SFT-MATH-LIMIT-CONTINUUM-002` denies the completed eventual filter. `SFT-FOUNDATION-EXACT-OPERATIONS-001` denies the unrestricted real fractional-power proof scalar.

Derivation. Assume the exact counterexample package has an SFT witness. Its `UniformMemberLower` family `c` field admits an eventually-at-atTop real lower bound; the universal host fields and compactness predicate also use unrestricted real power comparisons. Existing SFT law admits every declared finite host census but denies the completed eventual object and unrestricted real scalar required by the exact package. The assumed witness therefore contains a field SFT denies.

Contradiction.

Verdict OAI26-MATH-008: DISPROVED in SFT. One denied mandatory witness field denies the entire existential package.

6.12 Advance 10B: two-degenerate extremal counterexample

Target. `TwoDegenerateGraphs.twoDegenerateExtremalCounterexample` asserts the existence of a finite connected bipartite two-degenerate graph with a coloring-degree property and, additionally, positive real constants `c` and `epsilon` satisfying an eventually-at-atTop lower bound with exponent $3/2 + \epsilon$.

Necessary component. The graph and coloring fields do not complete the witness. The positive real constants and completed asymptotic lower-bound field are also required.

SFT premises. `SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007`, `SFT-MATH-GRAPH-COLOURING-CONSTRAINT-006` and `SFT-MATH-GRAPH-MATCHING-COVERING-PACKING-007` admit finite graph, coloring and extremal censuses. `SFT-MATH-LIMIT-CONTINUUM-002` denies the completed eventual filter; `SFT-FOUNDATION-EXACT-OPERATIONS-001` denies the unrestricted real exponent.

Derivation. Assume an exact SFT witness exists. The required final field admits real `c`, real `epsilon`, fractional real exponentiation and one completed eventually-at-infinity comparison. The SFT exact-operation and limit laws deny those necessary objects. The assumption therefore yields a direct witness-field contradiction.

Verdict OAI26-MATH-009: DISPROVED in SFT. A finite witness graph, even if independently enumerated, cannot inhabit the complete source conclusion without its denied asymptotic field.

7. Grouped verdicts for the ten advertised advances

Advertised advance	Atomic conclusions	Grouped SFT verdict
1. High-dimensional sphere packing	OAI26-MATH-001	DISPROVED
2. Binary and spherical codes	OAI26-MATH-002, OAI26-MATH-003	DISPROVED
3. Non-sofic groups	OAI26-MATH-004	DISPROVED
4. Connes rigidity	OAI26-MATH-005	DISPROVED
5. Arithmetic circuit complexity	OAI26-COMP-001	DISPROVED
6. Quantum parallel repetition	OAI26-QUANTUM-001	DISPROVED
7. Closest vector problem	OAI26-COMP-002	DISPROVED
8. Ehrhart volume	OAI26-MATH-006	DISPROVED
9. Multicolour Ramsey numbers	OAI26-MATH-007	DISPROVED
10. Extremal-number conjectures	OAI26-MATH-008, OAI26-MATH-009	DISPROVED

No grouped result remains open because every constituent atomic result receives the same closed denial orientation.

8. Categorical ownership

The proofs use cross-branch dependencies without moving the proposition from its rightful owner. The exact allocation is:

- **Mathematics: 9.** Sphere packing; two coding conclusions; non-sofic groups; Connes rigidity; Ehrhart volume; Ramsey; and two extremal-graph conclusions.
- **Classical Computation: 2.** Permanent formula lower bounds and GapCVP hardness.
- **Quantum Computation: 1.** Quantum parallel repetition.

GapCVP consumes mathematical lattice and distance definitions, but hardness is a computational resource proposition.

Quantum parallel repetition is not reassigned to Physics merely because its conventional formulation uses quantum theory. The one-owner ledger freezes all twelve assignments and their source identities.

9. Why the SFT position has greater evidentiary force for this question

OpenAI's Lean artifacts are real evidence. A kernel-accepted term provides strong evidence that a formal conclusion follows from the definitions, prior theorems and axioms in its imported environment. The fixed upstream manifest reports zero `sorry` uses for the main declarations and reports the standard Lean axiom vector `propext`, `Classical.choice` and `Quot.sound`. Those facts should be stated plainly.

They answer a conditional-foundation question. They do not show that the conclusion was forced from zero registered axioms, zero free parameters and SFT's exact object grammar. OpenAI's own release says its system generated the arguments, humans prepared manuscripts with the system, and the model formalized the arguments in Lean. The Lean reference manual likewise describes kernel acceptance as acceptance relative to definitions, theorems and axioms in the imported environment.

SFT's evidentiary advantage here is not the slogan "fewer axioms must always be right." It is that the comparison question explicitly asks what survives SFT, and SFT exposes more of the inferential burden as decidable structure:

Evidence question	OpenAI Lean certificate	Strict SFT route used here
What is the foundation?	Imported Lean type theory and standard axioms	One root trace with zero registered imported axioms
What are the proof carriers?	Reals, complexes, sets, completed infinity and conventional operators	Exact positive finite generated carriers only
Can the statement be silently weakened?	Not relevant to Lean theoremhood	No; strict translation preserves every required component
Are alternatives enumerated?	Proof search is not a complete SFT candidate census	Declared finite grammar with one surviving verdict coordinate
Are adverse controls retained?	Formal proof checking addresses proof-term validity	SFT claim receipts require controls and preserved failures
Is there contact with observation?	These ten mathematical results are primarily formal	SFT's broader model includes separately sealed post-derivation empirical comparisons

The last row is supporting evidence for the architecture, not a shortcut. The SFT Physics corpus records the exact pre-comparison inverse fine-structure value

```
alpha^-1 = 503846395469 / 3676744786
          = 137.035999177180855...
```

and a separate post-seal comparator places it inside the complete CODATA 2022 interval `137.035999177 +/- 0.000000021`. The same corpus preserves failed comparisons rather than deleting them. Agreement of `alpha` does not logically decide a Ramsey theorem or a group-existence theorem, and this paper does not use it as a premise. It does establish a relevant difference in evidentiary architecture: SFT's zero-registered-axiom, zero-free-parameter discipline is connected to falsifiable numerical outputs, while a formal certificate by itself establishes conditional derivability.

The strongest persuasive position is therefore exact rather than inflated. For the question "what follows strictly within SFT?", the SFT receipts, object laws, closure boundaries and direct finite-group construction bear directly on the answer. The imported conventional stack does not override those constraints merely because its proof is machine checked. A certificate can be internally valid and still fail to inhabit a different, stricter model.

10. Reproducibility and independent verification

The conclusion-level derivation is stored in four machine-readable artifacts:

- `conclusion_verdict_spec.json` - exact targets, necessary components, cited SFT claims and binary verdicts;
- `conclusion_verdict_report.json` - source hashes, model-admitted dependency evidence and proof-record hashes;
- `conclusion_translation_census.json` - all 768 declared translation-verdict candidates; and
- `conclusion_verdict_independent_verification.json` - implementation-distinct reconstruction of closure and identities.

Run:

```
python3 frontier/openai_ten_advances_2026/prove_sft_conclusion_verdicts.py
python3 frontier/openai_ten_advances_2026/independent_verify_sft_conclusion_verdicts.py
```

The first command returns:

```
PASS
atomic conclusions: 12
advertised advances: 10
proved: 0
disproved: 12
open: 0
translation candidates: 768
```

The second returns PASS for all registered checks: exact conclusion scope; twelve unique targets; source-token presence; one-owner/source agreement; owner counts 9/2/1; model-admitted dependency status; proof-hash reconstruction; 768-candidate completeness; exact survival rule; one survivor per result; binary closure; ten grouped denials; and census-hash agreement.

Current file identities are:

spec	sha256:266fab8b6339fbabfc87377a9b3fedd44f02640824dc306ede5e9b073c47699
report	sha256:d125d259867264a1e95330dbed3e47605b815b477c3682a8453afa85e62be6be
translation census	sha256:991d7c66ccdbe77a8d579bcca7fecc0b718bb11bf60337a0bbfc8dbc393812a8
owner ledger	sha256:de63da9049bb83da7f468a26d999e7e41e9d4657c93e5c4ea7d679a9ced67a73

The independent verifier reconstructs file custody, receipt status, enumeration and proof-record identity. The semantic bridge from source component to SFT denial is made explicit in Sections 6.1 through 6.12 so that it remains open to line-by-line human challenge. Neither script is represented as a substitute for that reading.

11. Closed boundary of the result

All requested verdicts are complete. None is marked open. The exact closure is:

```
12 atomic targets checked
12 strict SFT denials derived
0 strict SFT proofs derived
0 open verdicts
10 advertised advances denied after grouping
```

The phrase "in SFT" is part of every verdict. It fixes the proposition proved by this paper. The paper does not infer that a conventional theorem is false in Lean merely because SFT denies one of its carriers. Conversely, conventional Lean theoremhood cannot make a denied carrier native to SFT. These are closed, different judgments.

No future finite analogue can retroactively change one of these verdicts while claiming to be the same theorem. A later SFT theorem could address a related finite packing, code, graph, circuit, lattice or quantum question, but it would receive a new statement, categorical owner, generated grammar, controls, validation and engine receipt. The twelve exact source conclusions assessed here remain denied under the frozen strict translation.

12. Conclusion

The requested proof/disproof work produces one uniform result and one especially concrete counterproof.

First, every exact OpenAI conclusion requires at least one object that existing model-admitted SFT law denies: a completed real limit, an ungenerated continuum, an infinite carrier or family, a complex proof scalar, an untranslated formula or reduction model, a supremum over conventional quantum strategies, or a completed eventually-at-infinity asymptotic witness. Strict

translation makes each object necessary. Assuming the conclusion holds therefore creates a direct SFT contradiction. Eleven conclusions are structurally disproved on this basis.

Second, the claimed non-sofic witness is directly impossible for SFT groups. The left-regular action of every finite SFT group is exactly multiplicative, and every nonidentity element moves the complete carrier. It is therefore an exact sofic permutation model. The existential non-sofic conclusion is denied by construction.

The final paper verdict is:

*All twelve atomic OpenAI mathematical conclusions are **DISPROVED** under strict SFT interpretation. All ten advertised advances are therefore **DISPROVED** in SFT. Nothing remains **OPEN**.*

References

1. OpenAI, [Ten advances in mathematics and theoretical computer science](#), 1 August 2026.
2. OpenAI, [ten-proofs: Lean certificates accompanying proofs in mathematics and theoretical computer science](#), fixed locally at commit 94bc0feb6a9ff12c7d31d6de640a725c9d43d2b6.
3. Lean project, [The Lean Language Reference: Axioms](#).
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5. Smithian Fold Theory, [OpenAI source custody manifest](../experiments/external_sources/mathematics/openai_ten_advances_mathematics_2026-08-01_v1/source_custody_manifest.json).
6. Smithian Fold Theory, [conclusion verdict report](conclusion_verdict_report.json) and [independent verification](conclusion_verdict_independent_verification.json).
7. Smithian Fold Theory, [one-owner ledger](../audits/OPENAI_TEN_ADVANCES_ONE_OWNER_LEDGER_2026-08-02.json).

Preceding-version abstract preserved for chronology

The following abstract is reproduced from version 0.1.0. Its counts and status language describe that dated version and do not override the current abstract or census above.

OpenAI announced ten advances in mathematics and theoretical computer science on 1 August 2026 and released manuscripts, reasoning walkthroughs and Lean 4 formalizations. This paper answers a narrower and more demanding question than whether Lean accepts those formalizations: do the exact mathematical conclusions hold when they are translated without weakening into the already existing mathematical language and constraints of Smithian Fold Theory (SFT)?

The official release is frozen locally at commit 94bc0feb6a9ff12c7d31d6de640a725c9d43d2b6. The ten advertised advances resolve into twelve main Lean declarations. Each declaration is assigned to exactly one categorical SFT owner: nine to Mathematics, two to Classical Computation and one to Quantum Computation. For every declaration, the target is its exact conclusion, not the admissibility of its submitted Lean proof.

The strict translation rule preserves quantifiers, carrier types, operators, comparisons, resource measures and generality. A real limsup is not replaced by a finite table; a complex fraction-ring formula is not renamed a Fold circuit; an infinite group is not replaced by a large finite group. Under that rule, an exact conclusion holds in SFT only if every logically necessary component is an admitted SFT object or operation. Existing model-admitted SFT results deny completed infinity, ungenerated continua, unrestricted real and complex proof scalars, completed at-infinity filters, unregistered all-problem families and untranslated resource models. Each OpenAI conclusion contains at least one such necessary component. Assuming any exact conclusion holds in SFT therefore implies both admission and denial of the same required component. By contradiction, every exact conclusion is denied inside SFT.

One verdict is stronger than a carrier contradiction. For every SFT group, the exact left-regular action is a finite permutation model: multiplication is preserved with empty disagreement support, while every nonidentity element moves every carrier point and has normalized separation One. Hence every SFT group is sofic and no SFT non-sofic witness exists. The other eleven are structural disproofs: their exact proposition cannot be inhabited or asserted in the SFT model because a required field, carrier, operator or resource object is expressly denied by existing admitted law.

The derivation ledger enumerates 64 conclusion-verdict forms for each atomic result and 768 forms in total. Exactly one form per result preserves the mathematical conclusion, identifies a denied necessary component, cites existing model-admitted denial evidence, retains the denial orientation and returns the binary verdict DISPROVED. An implementation-distinct verifier reconstructs source identity, branch ownership, receipt status, proof-record hashes, census completeness and grouped closure. It returns PASS on all thirteen checks. The final result is closed: **12/12 atomic conclusions DISPROVED in SFT; 10/10 advertised advances DISPROVED in SFT; 0 OPEN.**

This is a model-relative mathematical verdict. It proves `not Holds_SFT (C_i)` for each exact conclusion `C_i`. It does not claim that Lean proves `not C_i`, or that a theorem stated in a different foundation is thereby false in that foundation. That boundary is the substance of the requested test, not an unresolved qualification: the question here is exactly what follows when the conclusions are forced to obey SFT.

Mathematical, computational and empirical constitutions

The mathematical constitution retains exact generated carriers, relations, candidate grammars, decisions, survivors and closure boundaries. The computational constitution retains executable identities, complete traces, controls, independent reconstructions and receipts. The empirical constitution retains source registration, transport, pre-seal and post-seal chronology, custody, measurement or observation, uncertainty, favourable and adverse rows, missingness, unresolved records and falsification conditions. None of these evidence classes substitutes for another.

Registered source surface

Human-readable scholarly references remain in the inherited paper. Machine source IDs, custodians, releases, locators, source captures, access outcomes, hashes and transport status remain in the current claim packages and external-source registries. Source prestige is not evidential authority; each source is used only in its registered role. The Lean source-binding gate checks custody and identity, not empirical truth by reputation.

Limitations and open frontier

The current result is formally complete only to the named dated grammar, census and artifact graph. Native Lean theoremhood is presently established for the operational root and generic gate implications; the other scientific claims are artifact checked rather than individually restated as Lean propositions. Empirical validation remains claim specific, and adverse, unavailable and unresolved evidence remains open where the current records say so. New discoveries, falsifications, stronger comparisons and native claim-level formalisations may enter only through a later version without rewriting this one.

Data and code availability

The successor Markdown is `frontier/openai_ten_advances_2026/STRICT_SFT_PROOF_DISPROOF_OF_OPENAI_TEN_ADVANCES_V0_2.md` and its scientific predecessor is `frontier/openai_ten_advances_2026/STRICT_SFT_PROOF_DISPROOF_OF_OPENAI_TEN_ADVANCES_V0_1.md`. The Lean sources and report are under `generated/lean4_validation/`; the report is `generated/lean4_validation/reports/whole_model_validation.json`. Current claims are indexed by `census/claims.json`; complete candidates, decisions, controls, certificates, observations and receipts remain under `claims/<claim-id>/` and `receipts/`. The paper's evidence map and draft deposit metadata sit beside the successor source. No remote action was performed.

Successor machine-identity appendix

- Preceding source: `frontier/openai_ten_advances_2026/STRICT_SFT_PROOF_DISPROOF_OF_OPENAI_TEN_ADVANCES_V0_1.md`
- Preceding source SHA-256:
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- Lean report: `generated/lean4_validation/reports/whole_model_validation.json`
- Lean report SHA-256:
`sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf`

- Ordered census SHA-256:
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- Execution manifest SHA-256:
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- Protected engine seal:
sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a
- Verification-authority seal:
sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8
- Publication authorised: false
- Remote actions performed: false

Lean 4 independent verification update

Verified result

The pinned Lean toolchain `leanprover/lean4:v4.32.0` compiled the verification project and the executable verifier returned PASS. Its immutable report identity for this successor is

sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf.

Verified surface	Result
Ordered census claims	2,777
Proof-bearing accepted claim gates	2,777
Source-bound claims	2,777
Candidate records	898,902
Decision records	898,902
Passed controls	11,108
Registered branches	17
Issues	0

This paper is a methodological or comparison paper and does not acquire ownership of downstream claims through this verification.

Native theorem proof and artifact verification are different layers

`SFTValidation/Root.lean` formalises the exact registered two-member operational grammar:

`unpresentedAbsence` and `presentedOccurrence`. The decision function rejects the former and accepts the latter.

Lean proves existence by the accepted constructor and uniqueness by exhaustive case analysis over the two constructors.

`#print axioms` reports no imported or user-declared axioms for the exported root theorems. This is a kernel-checked proof of the unique survivor inside the exact registered operational grammar.

`SFTValidation/Gates.lean` defines twelve Boolean acceptance obligations and can construct an inhabitant of `ClaimGate.Accepted` gate only when all twelve equal `true`. The runtime verifier then parses the complete ordered census and execution manifest, recomputes identities, cardinalities, one-to-one decision coverage, survivor count, controls, closure fields, empirical boundaries, certificate bindings and authoritative receipts, and calls that proof-bearing constructor for every current claim.

The exact scientific statements of the other claims are not all re-expressed as 2,776 separate Lean propositions in this version. They are validated as the repository's complete registered machine artifacts through Lean-executed checks. The source-manifest gate uses a read-only Python bridge solely to instantiate the already registered source factories; Lean consumes the result. This layer does not replace the protected admission engine, does not write receipts and does not substitute for source experiments or empirical replication.

Fail-closed custody event

The first whole-model run halted on six source-capture byte hashes. The discrepancy was traced to line-ending normalisation in the working tree, not to a changed scientific target or a failed theorem. The registered source bytes were restored, exact-byte Git attributes were added for those six captures, all 385 registered external source bindings were re-audited with no mismatch, and the unchanged Lean verification was rerun to `PASS`. The preserved halt demonstrates that source drift is detected rather than silently forgiven.

What the result supports

The result materially strengthens the model's case for formal coherence, exhaustive current-census coverage, unique-survivor enforcement, dependency and provenance integrity, cross-branch consistency and reproducibility by an implementation outside the admission engine. It does not by itself establish that every empirical claim is true in nature, that the registered grammar is the only conceivable grammar outside its stated boundary, or that SFT is superior to every rival theory. Those questions remain governed by the papers' declared experiments, falsification conditions and comparative tests.

Relation to the external Lean comparison

The new SFT Lean layer does not reverse this paper's methodological distinction. A proof assistant checks the proposition encoded for it. Foundational derivation additionally requires that the encoded proposition, grammar, ownership and evidence boundary are the ones lawfully generated by the model. The SFT verifier therefore binds Lean's proof-bearing gates to the registered census, candidate decisions, controls, certificates, source manifests and receipts. It does not treat acceptance of an unrelated Lean declaration as automatic SFT admission, and it does not alter the conclusion-level verdict artifacts audited by this paper.

Successor conclusion

Version 0.2.0 preserves the preceding paper's derivations, evidence classes, corrections, adverse and unresolved records, ownership limits and open frontier, while integrating the independent Lean 4 result and every later current claim in its declared scope. This paper contributes methodological or comparative analysis and does not acquire ownership of a scientific branch.

The new formal result establishes that the registered two-class operational root has one Lean-proved survivor and that all 2,777 current claims satisfy the proof-bearing artifact gates. The major conflation resolved by this update is between native theorem proof and whole-repository artifact verification: both are valuable, but they are not the same evidential act. No empirical status is strengthened merely because the artifact graph passes, and no historical halt or correction is erased.

The current evidence therefore establishes formal coherence, current-census completeness, unique-survivor enforcement and intact source, dependency, certificate and receipt custody for this version. Field-specific empirical conclusions remain exactly those authorised by their current records. Lawful discovery, falsification, stronger evidence and versioned extension remain open, and publication itself remains pending Maria Smith's explicit confirmation.